Fig. S1. Questionnaire Flow

(a) Each point is one questionnaire response; horizontal bar = median, vertical bar = IQR. (b) Participant-level mean in-game T17 score vs. questionnaire Flow.

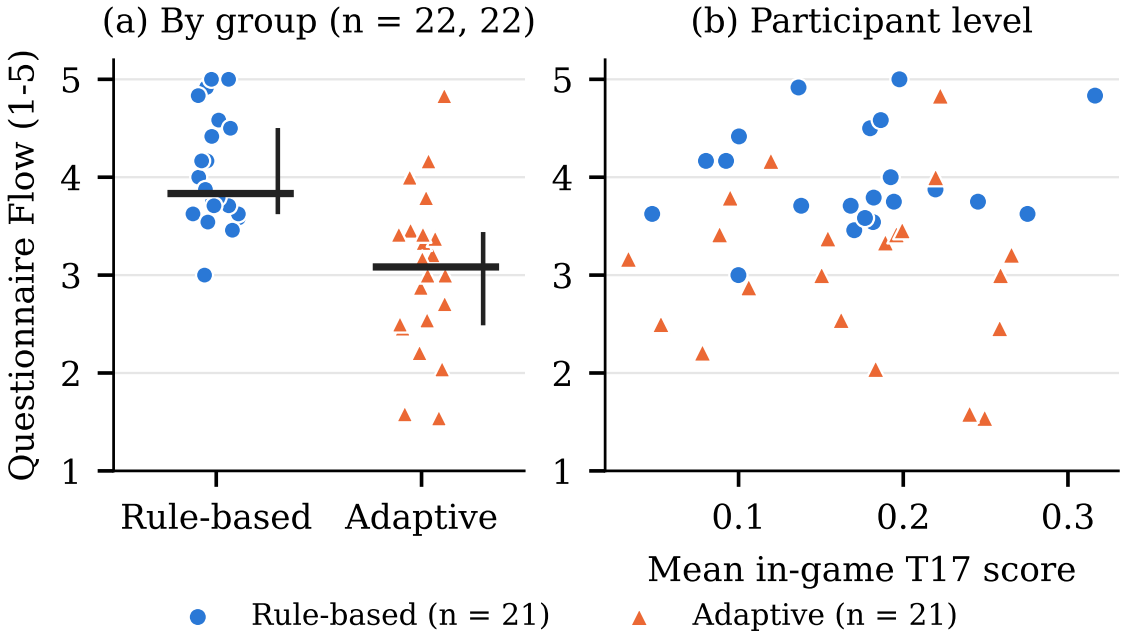


Fig. S2. Telemetry indicators vs. questionnaire Flow

Participant level, by condition. (a) Scatter plots with least-squares lines; (b) rank plots (ranks within condition). Used to inspect differences between Pearson r and Spearman ρ.

Fig. S2a. Participant-level scatter plots with least-squares lines (R = rule-based, A = adaptive)

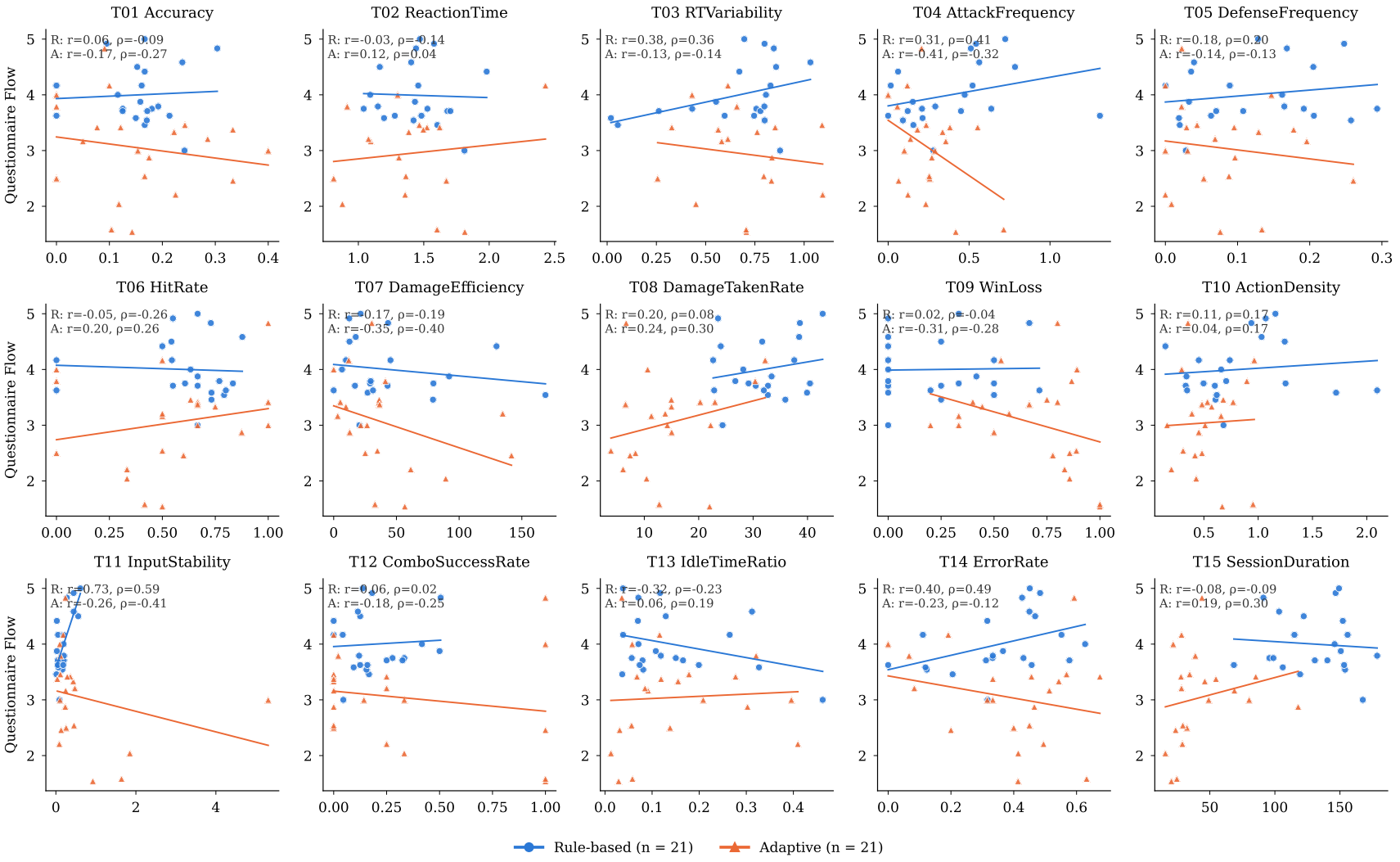


Fig. S2b. Rank plots (ranks within condition) of the same data

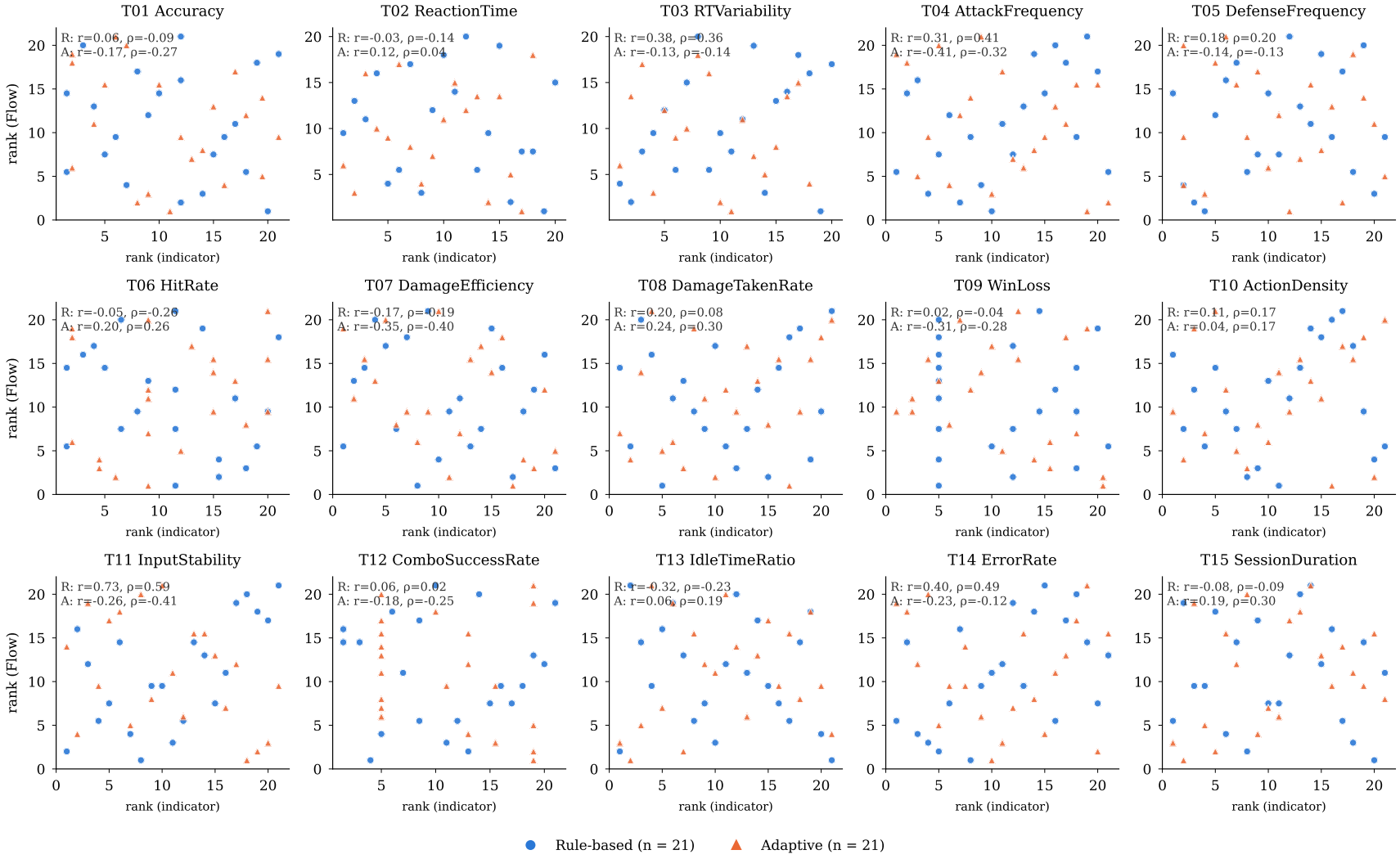


Fig. S3. Distributions

Participant-level histograms of questionnaire Flow (matched participants) and T01–T15 by condition; dashed lines = medians.

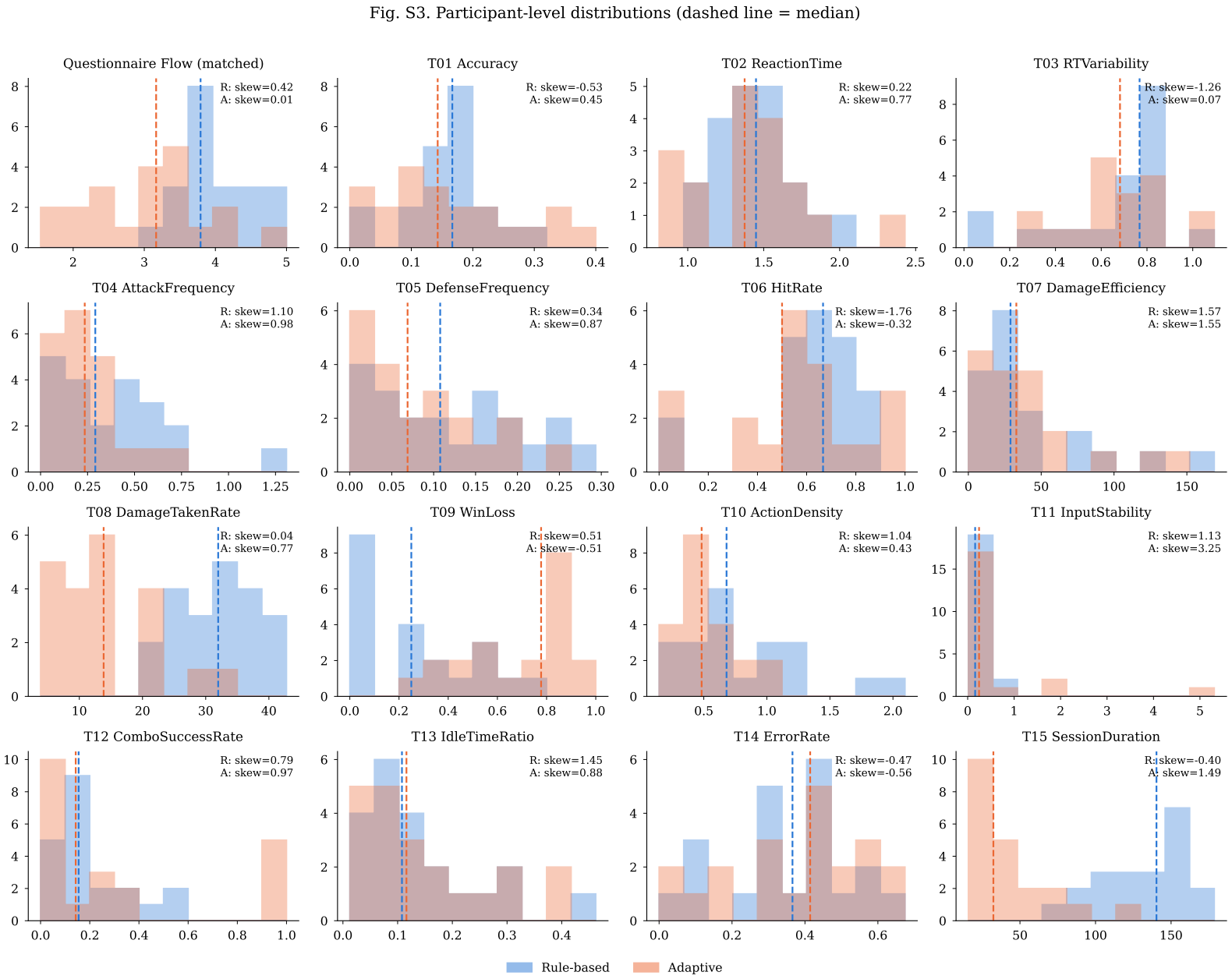


Table S1. Dataset summary by condition

Encounter = one boss fight; participant = one tester within one condition.

measure	Rule-based	Adaptive
participants_telemetry	24.000	21.000
questionnaire_responses	22.000	22.000
participants_matched	21.000	21.000
encounters	113.000	206.000
encounters_per_participant_median	4.000	7.000
min	2.000	2.000
max	12.000	37.000
duration_median_s	130.350	29.500
duration_q1	103.490	16.860
duration_q3	155.710	54.580
wins	28.000	157.000
win_pct	24.780	76.210
max_boss_level	5.000	31.000
T17_zero	8.000	25.000
reaction_samples_lt2	40.000	133.000
T11_gt50	0.000	8.000

Table S2. Questionnaire reliability (Cronbach alpha) by scale and condition

condition	scale	alpha	mean	sd	mean_inter_item_r
Rule-based	All 24 items	0.917	4.045	0.562	0.333
Rule-based	Concentration	0.781	4.148	0.591	0.474
Rule-based	Challenge-skill balance	0.585	3.875	0.797	0.267
Rule-based	Control / system response	0.845	3.943	0.824	0.575
Rule-based	Enjoyment / motivation	0.842	4.432	0.573	0.575
Rule-based	Time / deep engagement	0.732	4.000	0.744	0.444
Rule-based	Pressure / flow experience	0.795	3.875	0.801	0.545
Adaptive	All 24 items	0.932	3.028	0.809	0.369
Adaptive	Concentration	0.777	3.284	0.989	0.470
Adaptive	Challenge-skill balance	0.510	3.159	0.918	0.167
Adaptive	Control / system response	0.738	3.000	0.973	0.404
Adaptive	Enjoyment / motivation	0.876	3.125	1.023	0.641
Adaptive	Time / deep engagement	0.697	3.091	0.868	0.359
Adaptive	Pressure / flow experience	0.704	2.511	0.962	0.364

Table S3. Questionnaire item statistics

Floor/ceiling = % of responses at 1/5. Corrected item–total r uses the 24-item total; item–rest r and alpha if deleted use the item's dimension.

condition	dimension	item	mean	sd	floor_pct	ceiling_pct	corrected_item_total_r_scale	item_rest_r_dimension	alpha_if_deleted_dimension
Rule-based	Concentration	q1	4.273	0.631	0.000	36.364	0.689	0.454	0.788
Rule-based	Concentration	q2	4.182	0.733	0.000	36.364	0.422	0.657	0.692
Rule-based	Concentration	q3	3.864	0.889	0.000	22.727	0.471	0.596	0.730
Rule-based	Concentration	q4	4.273	0.767	0.000	45.455	0.513	0.664	0.686
Rule-based	Challenge-skill balance	q5	3.727	1.279	0.000	40.909	0.513	0.564	0.333
Rule-based	Challenge-skill balance	q6	3.909	1.306	4.545	50.000	0.525	0.150	0.688
Rule-based	Challenge-skill balance	q7	3.773	1.110	0.000	36.364	0.564	0.558	0.366

condition	dimension	item	mean	sd	floor_pct	ceiling_pct	corrected_item_total_r_scale	item_rest_r_dimension	alpha_if_deleted_dimension
Rule-based	Challenge-skill balance	q8	4.091	1.065	0.000	50.000	0.321	0.264	0.584
Rule-based	Control / system response	q9	3.773	1.110	0.000	36.364	0.564	0.760	0.767
Rule-based	Control / system response	q10	3.909	1.065	0.000	36.364	0.500	0.779	0.758
Rule-based	Control / system response	q11	4.000	0.926	0.000	36.364	0.737	0.510	0.870
Rule-based	Control / system response	q12	4.091	0.868	0.000	36.364	0.488	0.703	0.799
Rule-based	Enjoyment / motivation	q13	4.727	0.550	0.000	77.273	0.419	0.691	0.805
Rule-based	Enjoyment / motivation	q14	4.409	0.666	0.000	50.000	0.522	0.454	0.888
Rule-based	Enjoyment / motivation	q15	4.227	0.813	0.000	40.909	0.562	0.789	0.749
Rule-based	Enjoyment / motivation	q16	4.364	0.727	0.000	50.000	0.646	0.838	0.724
Rule-based	Time / deep engagement	q17	4.045	1.046	4.545	40.909	0.530	0.293	0.802
Rule-based	Time / deep engagement	q18	4.182	0.795	0.000	36.364	0.450	0.703	0.597
Rule-based	Time / deep engagement	q19	3.636	1.293	9.091	31.818	0.614	0.628	0.619
Rule-based	Time / deep engagement	q20	4.136	0.774	0.000	36.364	0.661	0.616	0.641
Rule-based	Pressure / flow experience	q21	4.000	1.113	4.545	40.909	0.681	0.560	0.771
Rule-based	Pressure / flow experience	q22	4.273	0.631	0.000	36.364	0.582	0.615	0.772
Rule-based	Pressure / flow experience	q23	3.318	1.287	4.545	27.273	0.588	0.558	0.797
Rule-based	Pressure / flow experience	q24	3.909	0.921	0.000	31.818	0.699	0.850	0.631
Adaptive	Concentration	q1	3.409	1.297	4.545	27.273	0.652	0.691	0.662
Adaptive	Concentration	q2	3.500	1.336	9.091	27.273	0.801	0.732	0.636
Adaptive	Concentration	q3	3.091	1.306	9.091	18.182	0.346	0.255	0.879
Adaptive	Concentration	q4	3.136	1.167	4.545	13.636	0.787	0.712	0.660
Adaptive	Challenge-skill balance	q5	3.045	1.588	22.727	31.818	0.703	0.643	0.041
Adaptive	Challenge-skill balance	q6	2.591	1.623	40.909	18.182	0.337	0.434	0.297
Adaptive	Challenge-skill balance	q7	2.818	1.332	27.273	9.091	0.329	0.422	0.336
Adaptive	Challenge-skill balance	q8	4.182	1.181	4.545	54.545	0.115	-0.194	0.746
Adaptive	Control / system response	q9	2.682	1.287	13.636	13.636	0.684	0.730	0.557
Adaptive	Control / system response	q10	2.727	1.420	18.182	18.182	0.655	0.604	0.632
Adaptive	Control / system response	q11	3.364	1.293	9.091	27.273	0.684	0.607	0.633
Adaptive	Control / system response	q12	3.227	1.193	9.091	13.636	0.418	0.222	0.826
Adaptive	Enjoyment / motivation	q13	3.455	1.184	4.545	18.182	0.728	0.787	0.820
Adaptive	Enjoyment / motivation	q14	3.409	1.054	4.545	13.636	0.764	0.672	0.866
Adaptive	Enjoyment / motivation	q15	2.682	1.287	18.182	13.636	0.759	0.721	0.848
Adaptive	Enjoyment / motivation	q16	2.955	1.253	13.636	13.636	0.646	0.767	0.828
Adaptive	Time / deep engagement	q17	3.409	1.141	9.091	13.636	0.219	0.182	0.795
Adaptive	Time / deep engagement	q18	3.091	1.231	9.091	13.636	0.662	0.645	0.521
Adaptive	Time / deep engagement	q19	2.591	1.221	22.727	9.091	0.652	0.421	0.671
Adaptive	Time / deep engagement	q20	3.273	1.202	9.091	18.182	0.820	0.740	0.455
Adaptive	Pressure / flow experience	q21	2.318	1.211	36.364	4.545	0.654	0.377	0.704
Adaptive	Pressure / flow experience	q22	3.318	1.211	4.545	22.727	0.330	0.349	0.718
Adaptive	Pressure / flow experience	q23	2.227	1.510	50.000	13.636	0.702	0.641	0.533
Adaptive	Pressure / flow experience	q24	2.182	1.332	40.909	9.091	0.673	0.614	0.560

Table S4. Descriptive statistics of questionnaire Flow (mean of 24 items)

Outliers: Tukey rule (beyond 1.5 IQR).

condition	n	mean	sd	median	q1	q3	min	max	skewness	kurtosis_excess	shapiro_p	missing_pct	outliers_tukey_n
Rule-based	22	4.045	0.562	3.833	3.646	4.479	3.000	5.000	0.365	-0.842	0.103	0.000	0
Adaptive	22	3.028	0.809	3.083	2.510	3.417	1.542	4.833	0.064	-0.144	0.942	0.000	1

Table S5. Descriptive statistics of telemetry indicators T01–T15

Participant level = median per participant (T09 = win rate); encounter level = all encounters. Missing = undefined values (T02/T03 with fewer than 2 reaction samples).

condition	level	indicator	unit	n	mean	sd	median	q1	q3	min	max	skewness	kurtosis_excess	shapiro_p	missing_pct	outliers_tukey_n
Rule-based	participant	T01_Accuracy	ratio 0-1	21	0.157	0.070	0.167	0.125	0.180	0.000	0.304	-0.529	0.950	0.078	0.000	3
Rule-based	encounter	T01_Accuracy	ratio 0-1	113	0.160	0.118	0.154	0.095	0.208	0.000	0.625	1.302	2.984	0.000	0.000	5
Rule-based	participant	T02_ReactionTime	s	20	1.446	0.245	1.449	1.258	1.585	1.041	1.981	0.216	-0.401	0.781	4.762	0
Rule-based	encounter	T02_ReactionTime	s	73	1.411	0.433	1.403	1.124	1.604	0.408	2.723	0.350	0.396	0.455	35.398	1
Rule-based	participant	T03_RTVariability	s	20	0.661	0.272	0.768	0.586	0.812	0.019	1.032	-1.262	0.648	0.003	4.762	2
Rule-based	encounter	T03_RTVariability	s	73	0.693	0.300	0.757	0.553	0.881	0.019	1.359	-0.615	0.215	0.000	35.398	4
Rule-based	participant	T04_AttackFrequency	attacks/s	21	0.386	0.318	0.291	0.152	0.543	0.000	1.308	1.102	1.303	0.040	0.000	1
Rule-based	encounter	T04_AttackFrequency	attacks/s	113	0.450	0.476	0.322	0.125	0.654	0.000	3.137	2.297	8.312	0.000	0.000	4
Rule-based	participant	T05_DefenseFrequency	actions/s	21	0.120	0.090	0.108	0.036	0.192	0.000	0.293	0.343	-1.153	0.127	0.000	0
Rule-based	encounter	T05_DefenseFrequency	actions/s	113	0.510	4.224	0.083	0.028	0.180	0.000	45.000	10.478	107.867	0.000	0.000	3
Rule-based	participant	T06_HitRate	ratio 0-1	21	0.612	0.229	0.667	0.550	0.732	0.000	0.878	-1.760	2.618	0.000	0.000	2
Rule-based	encounter	T06_HitRate	ratio 0-1	113	0.599	0.283	0.667	0.524	0.750	0.000	1.000	-1.034	0.305	0.000	0.000	16
Rule-based	participant	T07_DamageEfficiency	damage/attack	21	43.601	43.364	28.934	16.975	45.276	0.000	168.818	1.574	1.754	0.001	0.000	3
Rule-based	encounter	T07_DamageEfficiency	damage/attack	113	104.020	332.850	22.208	9.766	66.723	0.000	2400	5.601	31.670	0.000	0.000	13
Rule-based	participant	T08_DamageTakenRate	damage/s	21	31.823	6.327	31.974	26.739	37.566	22.653	42.786	0.044	-1.181	0.281	0.000	0
Rule-based	encounter	T08_DamageTakenRate	damage/s	113	67.217	363.940	31.974	25.799	39.202	0.000	3900	10.474	107.811	0.000	0.000	4
Rule-based	participant	T09_WinLoss	win rate (0-1)	21	0.234	0.243	0.250	0.000	0.417	0.000	0.714	0.505	-1.009	0.004	0.000	0
Rule-based	encounter	T09_WinLoss	win rate (0-1)	113	0.239	0.428	0.000	0.000	0.000	0.000	1.000	1.224	-0.501	0.000	0.000	27
Rule-based	participant	T10_ActionDensity	actions/s	21	0.821	0.481	0.680	0.497	1.070	0.145	2.096	1.040	0.719	0.069	0.000	1
Rule-based	encounter	T10_ActionDensity	actions/s	113	1.423	5.609	0.679	0.443	1.186	0.000	60.000	10.214	104.164	0.000	0.000	8
Rule-based	participant	T11_InputStability	1/variance (s^-2)	21	0.195	0.177	0.161	0.055	0.200	0.015	0.613	1.132	0.198	0.003	0.000	4
Rule-based	encounter	T11_InputStability	1/variance (s^-2)	113	0.290	0.391	0.127	0.041	0.420	0.000	2.201	2.217	5.770	0.000	0.000	9
Rule-based	participant	T12_ComboSuccessRate	ratio 0-1	21	0.194	0.149	0.155	0.114	0.278	0.000	0.505	0.791	-0.309	0.049	0.000	0
Rule-based	encounter	T12_ComboSuccessRate	ratio 0-1	113	0.239	0.252	0.146	0.053	0.333	0.000	1.000	1.341	1.293	0.000	0.000	5
Rule-based	participant	T13_IdleTimeRatio	ratio 0-1	21	0.141	0.113	0.108	0.071	0.166	0.037	0.462	1.448	1.336	0.001	0.000	3
Rule-based	encounter	T13_IdleTimeRatio	ratio 0-1	113	0.165	0.166	0.117	0.070	0.204	0.003	1.000	2.728	9.784	0.000	0.000	9
Rule-based	participant	T14_ErrorRate	ratio 0-1	21	0.355	0.167	0.366	0.312	0.457	0.000	0.627	-0.474	-0.532	0.371	0.000	1
Rule-based	encounter	T14_ErrorRate	ratio 0-1	113	0.362	0.212	0.373	0.216	0.514	0.000	0.830	-0.103	-0.698	0.010	0.000	0
Rule-based	participant	T15_SessionDuration	s	21	130.725	28.803	140.340	106.090	152.210	68.490	178.590	-0.402	-0.738	0.376	0.000	0
Rule-based	encounter	T15_SessionDuration	s	113	130.196	48.432	130.350	103.490	155.710	1.000	419.960	1.728	10.138	0.000	0.000	3
Adaptive	participant	T01_Accuracy	ratio 0-1	21	0.159	0.114	0.143	0.091	0.225	0.000	0.400	0.448	-0.598	0.395	0.000	0
Adaptive	encounter	T01_Accuracy	ratio 0-1	206	0.166	0.198	0.116	0.000	0.250	0.000	1.000	2.013	5.541	0.000	0.000	5
Adaptive	participant	T02_ReactionTime	s	18	1.396	0.383	1.374	1.146	1.583	0.811	2.430	0.768	1.172	0.229	14.286	1
Adaptive	encounter	T02_ReactionTime	s	73	1.397	0.523	1.385	1.025	1.669	0.554	2.989	0.613	0.531	0.026	64.563	2
Adaptive	participant	T03_RTVariability	s	18	0.676	0.229	0.682	0.568	0.824	0.256	1.094	0.075	-0.392	0.822	14.286	0
Adaptive	encounter	T03_RTVariability	s	73	0.612	0.334	0.681	0.317	0.869	0.007	1.415	0.050	-0.928	0.037	64.563	0
Adaptive	participant	T04_AttackFrequency	attacks/s	21	0.252	0.171	0.235	0.122	0.336	0.000	0.714	0.977	0.898	0.176	0.000	1
Adaptive	encounter	T04_AttackFrequency	attacks/s	206	0.280	0.320	0.197	0.030	0.381	0.000	1.880	1.765	4.017	0.000	0.000	12
Adaptive	participant	T05_DefenseFrequency	actions/s	21	0.080	0.072	0.069	0.023	0.129	0.000	0.260	0.869	0.016	0.062	0.000	0
Adaptive	encounter	T05_DefenseFrequency	actions/s	206	0.092	0.102	0.063	0.000	0.145	0.000	0.607	1.471	2.906	0.000	0.000	4
Adaptive	participant	T06_HitRate	ratio 0-1	21	0.545	0.303	0.500	0.417	0.667	0.000	1.000	-0.319	-0.431	0.114	0.000	3
Adaptive	encounter	T06_HitRate	ratio 0-1	206	0.468	0.400	0.500	0.000	0.800	0.000	1.000	0.034	-1.523	0.000	0.000	0
Adaptive	participant	T07_DamageEfficiency	damage/attack	21	40.414	38.755	32.944	12.795	41.201	0.000	141.750	1.551	1.632	0.001	0.000	3
Adaptive	encounter	T07_DamageEfficiency	damage/attack	206	116.594	381.693	30.673	0.646	92.092	0.000	4123	8.238	76.023	0.000	0.000	19
Adaptive	participant	T08_DamageTakenRate	damage/s	21	14.631	7.838	13.851	8.389	20.278	3.869	32.213	0.770	-0.202	0.107	0.000	0
Adaptive	encounter	T08_DamageTakenRate	damage/s	206	21.769	22.404	15.258	5.895	31.483	0.000	144.912	1.907	5.184	0.000	0.000	9
Adaptive	participant	T09_WinLoss	win rate (0-1)	21	0.681	0.238	0.778	0.500	0.857	0.200	1.000	-0.509	-0.977	0.086	0.000	0
Adaptive	encounter	T09_WinLoss	win rate (0-1)	206	0.767	0.424	1.000	1.000	1.000	0.000	1.000	-1.263	-0.405	0.000	0.000	48
Adaptive	participant	T10_ActionDensity	actions/s	21	0.532	0.229	0.483	0.390	0.669	0.161	0.964	0.426	-0.598	0.454	0.000	0
Adaptive	encounter	T10_ActionDensity	actions/s	206	0.613	0.463	0.483	0.275	0.890	0.012	2.572	1.221	1.510	0.000	0.000	4
Adaptive	participant	T11_InputStability	1/variance (s^-2)	21	0.646	1.174	0.245	0.113	0.453	0.036	5.308	3.248	10.183	0.000	0.000	3
Adaptive	encounter	T11_InputStability	1/variance (s^-2)	206	24.450	172.780	0.209	0.035	1.151	0.000	1953	9.300	91.536	0.000	0.000	35
Adaptive	participant	T12_ComboSuccessRate	ratio 0-1	21	0.313	0.410	0.143	0.000	0.333	0.000	1.000	0.972	-0.767	0.000	0.000	5
Adaptive	encounter	T12_ComboSuccessRate	ratio 0-1	206	0.401	0.445	0.174	0.000	1.000	0.000	1.000	0.468	-1.597	0.000	0.000	0
Adaptive	participant	T13_IdleTimeRatio	ratio 0-1	21	0.150	0.123	0.116	0.058	0.209	0.013	0.409	0.884	-0.470	0.011	0.000	0
Adaptive	encounter	T13_IdleTimeRatio	ratio 0-1	206	0.154	0.175	0.091	0.029	0.229	0.000	0.953	1.652	2.756	0.000	0.000	9
Adaptive	participant	T14_ErrorRate	ratio 0-1	21	0.388	0.189	0.414	0.316	0.513	0.000	0.674	-0.560	-0.586	0.315	0.000	1
Adaptive	encounter	T14_ErrorRate	ratio 0-1	206	0.322	0.249	0.333	0.041	0.510	0.000	0.875	0.082	-1.193	0.000	0.000	0
Adaptive	participant	T15_SessionDuration	s	21	42.910	25.782	32.410	27.830	49.130	15.845	117.895	1.491	1.602	0.002	0.000	2
Adaptive	encounter	T15_SessionDuration	s	206	49.837	77.088	29.505	16.863	54.583	3.660	899.800	7.197	71.304	0.000	0.000	19

Table S6. Correlations of telemetry indicators with questionnaire Flow (step 3A)

Participant level, within condition. 95% CI by Fisher z (Bonett–Wright SE for Spearman). q = Benjamini–Hochberg FDR within each family of 15 tests (condition x method); T17 not included.

condition	indicator	n	pearson_r	pearson_ci_low	pearson_ci_high	pearson_p	spearman_rho	spearman_ci_low	spearman_ci_high	spearman_p	pearson_q_BH	spearman_q_BH
Rule-based	T01_Accuracy	21	0.057	-0.385	0.477	0.808	-0.086	-0.500	0.360	0.711	0.925	0.835
Rule-based	T02_ReactionTime	20	-0.034	-0.469	0.415	0.888	-0.138	-0.549	0.327	0.562	0.925	0.835
Rule-based	T03_RTVariability	20	0.381	-0.074	0.705	0.098	0.364	-0.109	0.702	0.115	0.488	0.432
Rule-based	T04_AttackFrequency	21	0.307	-0.144	0.652	0.176	0.408	-0.047	0.723	0.066	0.527	0.331
Rule-based	T05_DefenseFrequency	21	0.181	-0.272	0.568	0.432	0.198	-0.260	0.583	0.390	0.878	0.762
Rule-based	T06_HitRate	21	-0.053	-0.474	0.388	0.820	-0.262	-0.628	0.198	0.250	0.925	0.751
Rule-based	T07_DamageEfficiency	21	-0.167	-0.559	0.285	0.468	-0.191	-0.578	0.266	0.407	0.878	0.762
Rule-based	T08_DamageTakenRate	21	0.197	-0.256	0.579	0.392	0.082	-0.363	0.497	0.724	0.878	0.835
Rule-based	T09_WinLoss	21	0.022	-0.414	0.449	0.925	-0.037	-0.461	0.401	0.873	0.925	0.936
Rule-based	T10_ActionDensity	21	0.113	-0.335	0.520	0.624	0.168	-0.287	0.562	0.466	0.925	0.776
Rule-based	T11_InputStability	21	0.725	0.427	0.881	0.000	0.593	0.179	0.828	0.005	0.003	0.069
Rule-based	T12_ComboSuccessRate	21	0.063	-0.379	0.482	0.785	0.016	-0.418	0.445	0.944	0.925	0.944
Rule-based	T13_IdleTimeRatio	21	-0.324	-0.663	0.126	0.152	-0.229	-0.605	0.231	0.318	0.527	0.762
Rule-based	T14_ErrorRate	21	0.405	-0.032	0.712	0.069	0.486	0.043	0.770	0.025	0.488	0.190
Rule-based	T15_SessionDuration	21	-0.079	-0.494	0.365	0.733	-0.090	-0.503	0.357	0.699	0.925	0.835
Rule-based	T17_mean	21	0.175	-0.278	0.564	0.448	0.181	-0.276	0.571	0.433		
Adaptive	T01_Accuracy	21	-0.174	-0.563	0.279	0.450	-0.270	-0.634	0.191	0.236	0.676	0.454
Adaptive	T02_ReactionTime	18	0.123	-0.365	0.558	0.628	0.040	-0.435	0.498	0.874	0.724	0.874
Adaptive	T03_RTVariability	18	-0.135	-0.566	0.355	0.594	-0.144	-0.574	0.349	0.570	0.724	0.658
Adaptive	T04_AttackFrequency	21	-0.410	-0.715	0.026	0.065	-0.319	-0.666	0.142	0.159	0.676	0.454
Adaptive	T05_DefenseFrequency	21	-0.140	-0.539	0.310	0.545	-0.132	-0.535	0.320	0.568	0.724	0.658
Adaptive	T06_HitRate	21	0.205	-0.249	0.585	0.374	0.261	-0.200	0.627	0.254	0.676	0.454
Adaptive	T07_DamageEfficiency	21	-0.353	-0.681	0.093	0.117	-0.402	-0.719	0.055	0.071	0.676	0.454
Adaptive	T08_DamageTakenRate	21	0.242	-0.212	0.610	0.290	0.302	-0.159	0.655	0.183	0.676	0.454
Adaptive	T09_WinLoss	21	-0.310	-0.654	0.140	0.171	-0.276	-0.637	0.185	0.226	0.676	0.454
Adaptive	T10_ActionDensity	21	0.038	-0.400	0.462	0.870	0.172	-0.284	0.564	0.457	0.870	0.623
Adaptive	T11_InputStability	21	-0.262	-0.623	0.191	0.251	-0.411	-0.725	0.045	0.064	0.676	0.454
Adaptive	T12_ComboSuccessRate	21	-0.179	-0.567	0.274	0.437	-0.251	-0.620	0.210	0.273	0.676	0.454
Adaptive	T13_IdleTimeRatio	21	0.059	-0.383	0.478	0.800	0.190	-0.267	0.578	0.408	0.857	0.613
Adaptive	T14_ErrorRate	21	-0.228	-0.601	0.226	0.320	-0.117	-0.523	0.333	0.615	0.676	0.658
Adaptive	T15_SessionDuration	21	0.195	-0.258	0.578	0.397	0.303	-0.158	0.656	0.181	0.676	0.454
Adaptive	T17_mean	21	-0.097	-0.508	0.349	0.675	-0.072	-0.489	0.372	0.756		

Table S7. Spearman correlation matrix among T01–T15 (rule-based)

Pearson matrices are provided as CSV files in results/.

	T01	T02	T03	T04	T05	T06	T07	T08	T09	T10	T11	T12	T13	T14	T15
T01	1.000	0.245	0.185	0.322	-0.157	0.656	0.301	0.464	0.155	0.293	0.159	0.022	0.142	0.020	-0.226
T02	0.245	1.000	-0.117	-0.084	-0.317	-0.063	0.165	-0.018	-0.095	-0.200	-0.295	-0.114	-0.194	-0.035	0.095
T03	0.185	-0.117	1.000	0.287	0.323	0.008	-0.376	-0.257	-0.142	0.202	0.568	-0.209	0.147	0.322	0.239
T04	0.322	-0.084	0.287	1.000	0.122	0.397	-0.438	0.449	-0.016	0.842	0.783	0.147	0.257	0.790	-0.530
T05	-0.157	-0.317	0.323	0.122	1.000	-0.011	-0.091	-0.191	0.307	0.114	0.374	0.383	-0.304	0.087	0.056
T06	0.656	-0.063	0.008	0.397	-0.011	1.000	0.217	0.591	0.137	0.515	0.224	0.224	0.187	0.101	-0.348
T07	0.301	0.165	-0.376	-0.438	-0.091	0.217	1.000	0.100	0.567	-0.366	-0.475	0.125	-0.313	-0.506	0.027
T08	0.464	-0.018	-0.257	0.449	-0.191	0.591	0.100	1.000	0.207	0.482	0.225	0.257	-0.021	0.175	-0.590
T09	0.155	-0.095	-0.142	-0.016	0.307	0.137	0.567	0.207	1.000	0.015	-0.097	0.403	-0.372	-0.227	-0.285
T10	0.293	-0.200	0.202	0.842	0.114	0.515	-0.366	0.482	0.015	1.000	0.638	-0.057	0.303	0.429	-0.474
T11	0.159	-0.295	0.568	0.783	0.374	0.224	-0.475	0.225	-0.097	0.638	1.000	0.081	0.229	0.700	-0.192
T12	0.022	-0.114	-0.209	0.147	0.383	0.224	0.125	0.257	0.403	-0.057	0.081	1.000	-0.231	0.298	-0.391
T13	0.142	-0.194	0.147	0.257	-0.304	0.187	-0.313	-0.021	-0.372	0.303	0.229	-0.231	1.000	0.229	-0.170
T14	0.020	-0.035	0.322	0.790	0.087	0.101	-0.506	0.175	-0.227	0.429	0.700	0.298	0.229	1.000	-0.335

	T01	T02	T03	T04	T05	T06	T07	T08	T09	T10	T11	T12	T13	T14	T15
T15	-0.226	0.095	0.239	-0.530	0.056	-0.348	0.027	-0.590	-0.285	-0.474	-0.192	-0.391	-0.170	-0.335	1.000

Table S7. Spearman correlation matrix among T01–T15 (adaptive)

Pearson matrices are provided as CSV files in results/.

	T01	T02	T03	T04	T05	T06	T07	T08	T09	T10	T11	T12	T13	T14	T15
T01	1.000	0.297	0.443	-0.163	0.070	0.471	0.385	-0.225	-0.374	-0.363	0.030	-0.036	0.205	-0.000	0.194
T02	0.297	1.000	0.375	0.267	0.131	0.504	0.005	0.309	-0.030	0.424	-0.007	0.138	-0.203	0.197	-0.172
T03	0.443	0.375	1.000	0.086	-0.179	0.297	0.333	0.053	0.036	0.044	-0.158	0.079	0.166	0.294	-0.015
T04	-0.163	0.267	0.086	1.000	0.248	0.259	-0.204	0.240	-0.051	0.474	0.412	-0.181	-0.118	0.694	0.116
T05	0.070	0.131	-0.179	0.248	1.000	0.049	-0.194	-0.257	0.007	0.052	-0.099	-0.074	-0.295	0.118	0.057
T06	0.471	0.504	0.297	0.259	0.049	1.000	-0.249	0.178	-0.665	-0.071	0.105	-0.158	0.213	0.394	0.624
T07	0.385	0.005	0.333	-0.204	-0.194	-0.249	1.000	-0.238	0.388	-0.095	0.025	0.367	-0.270	-0.205	-0.504
T08	-0.225	0.309	0.053	0.240	-0.257	0.178	-0.238	1.000	-0.285	0.674	0.088	-0.145	0.290	-0.121	0.121
T09	-0.374	-0.030	0.036	-0.051	0.007	-0.665	0.388	-0.285	1.000	0.016	0.095	0.463	-0.509	-0.087	-0.727
T10	-0.363	0.424	0.044	0.474	0.052	-0.071	-0.095	0.674	0.016	1.000	0.005	-0.276	0.010	0.186	-0.039
T11	0.030	-0.007	-0.158	0.412	-0.099	0.105	0.025	0.088	0.095	0.005	1.000	0.208	-0.318	0.215	-0.171
T12	-0.036	0.138	0.079	-0.181	-0.074	-0.158	0.367	-0.145	0.463	-0.276	0.208	1.000	-0.495	-0.184	-0.610
T13	0.205	-0.203	0.166	-0.118	-0.295	0.213	-0.270	0.290	-0.509	0.010	-0.318	-0.495	1.000	0.051	0.703
T14	-0.000	0.197	0.294	0.694	0.118	0.394	-0.205	-0.121	-0.087	0.186	0.215	-0.184	0.051	1.000	0.292
T15	0.194	-0.172	-0.015	0.116	0.057	0.624	-0.504	0.121	-0.727	-0.039	-0.171	-0.610	0.703	0.292	1.000

Table S8. Variance inflation factors (complete cases)

condition	indicator	VIF	n_complete
Rule-based	T01_Accuracy	7.203	20
Rule-based	T02_ReactionTime	2.429	20
Rule-based	T03_RTVariability	3.837	20
Rule-based	T04_AttackFrequency	27.591	20
Rule-based	T05_DefenseFrequency	3.671	20
Rule-based	T06_HitRate	19.904	20
Rule-based	T07_DamageEfficiency	6.466	20
Rule-based	T08_DamageTakenRate	6.849	20
Rule-based	T09_WinLoss	7.540	20
Rule-based	T10_ActionDensity	12.558	20
Rule-based	T11_InputStability	5.147	20
Rule-based	T12_ComboSuccessRate	6.070	20
Rule-based	T13_IdleTimeRatio	4.353	20
Rule-based	T14_ErrorRate	13.942	20
Rule-based	T15_SessionDuration	3.707	20
Adaptive	T01_Accuracy	13.503	18
Adaptive	T02_ReactionTime	6.959	18
Adaptive	T03_RTVariability	2.785	18
Adaptive	T04_AttackFrequency	9.727	18
Adaptive	T05_DefenseFrequency	13.976	18
Adaptive	T06_HitRate	41.414	18
Adaptive	T07_DamageEfficiency	4.898	18
Adaptive	T08_DamageTakenRate	32.659	18
Adaptive	T09_WinLoss	4.714	18
Adaptive	T10_ActionDensity	23.666	18
Adaptive	T11_InputStability	5.851	18
Adaptive	T12_ComboSuccessRate	13.414	18

condition	indicator	VIF	n_complete
Adaptive	T13_IdleTimeRatio	7.542	18
Adaptive	T14_ErrorRate	12.956	18
Adaptive	T15_SessionDuration	10.390	18

Table S9. Multiple linear regression of Flow on T01–T15: model fit (step 3B)

Standardized variables, complete cases. With 4 and 2 residual degrees of freedom the models are overfitted and are reported for completeness only.

condition	n_complete_cases	predictors	df_resid	R2	adj_R2	F	model_p	max_VIF	condition_number
Rule-based	20	15	4.000	0.942	0.725	4.347	0.083	27.591	235.528
Adaptive	18	15	2.000	1.000	0.998	532.473	0.002	41.414	417.333

Table S10. Multiple linear regression coefficients (standardized beta)

Not interpretable because of overfitting and collinearity; see Table S9.

condition	indicator	beta_std	SE	t	p	ci_low	ci_high	VIF
Rule-based	T01_Accuracy	0.276	0.323	0.854	0.441	-0.620	1.171	7.203
Rule-based	T02_ReactionTime	0.173	0.187	0.922	0.409	-0.347	0.693	2.429
Rule-based	T03_RTVariability	0.399	0.235	1.692	0.166	-0.255	1.052	3.837
Rule-based	T04_AttackFrequency	-1.285	0.631	-2.035	0.112	-3.038	0.468	27.591
Rule-based	T05_DefenseFrequency	-0.451	0.230	-1.956	0.122	-1.090	0.189	3.671
Rule-based	T06_HitRate	-0.417	0.536	-0.778	0.480	-1.906	1.072	19.904
Rule-based	T07_DamageEfficiency	0.194	0.306	0.635	0.560	-0.655	1.043	6.466
Rule-based	T08_DamageTakenRate	-0.329	0.315	-1.047	0.354	-1.203	0.544	6.849
Rule-based	T09_WinLoss	-0.221	0.330	-0.670	0.539	-1.138	0.695	7.540
Rule-based	T10_ActionDensity	1.084	0.426	2.545	0.064	-0.099	2.267	12.558
Rule-based	T11_InputStability	1.082	0.273	3.967	0.017	0.325	1.839	5.147
Rule-based	T12_ComboSuccessRate	0.306	0.296	1.032	0.361	-0.517	1.128	6.070
Rule-based	T13_IdleTimeRatio	-0.733	0.251	-2.923	0.043	-1.430	-0.037	4.353
Rule-based	T14_ErrorRate	0.295	0.449	0.657	0.547	-0.951	1.541	13.942
Rule-based	T15_SessionDuration	-0.424	0.231	-1.831	0.141	-1.066	0.219	3.707
Adaptive	T01_Accuracy	-0.671	0.041	-16.324	0.004	-0.848	-0.494	13.503
Adaptive	T02_ReactionTime	-0.813	0.030	-27.563	0.001	-0.940	-0.686	6.959
Adaptive	T03_RTVariability	-0.207	0.019	-11.066	0.008	-0.287	-0.126	2.785
Adaptive	T04_AttackFrequency	-0.698	0.035	-20.012	0.002	-0.848	-0.548	9.727
Adaptive	T05_DefenseFrequency	-0.729	0.042	-17.420	0.003	-0.909	-0.549	13.976
Adaptive	T06_HitRate	2.432	0.072	33.782	0.001	2.123	2.742	41.414
Adaptive	T07_DamageEfficiency	-0.771	0.025	-31.147	0.001	-0.878	-0.665	4.898
Adaptive	T08_DamageTakenRate	-1.473	0.064	-23.036	0.002	-1.748	-1.198	32.659
Adaptive	T09_WinLoss	-0.332	0.024	-13.676	0.005	-0.437	-0.228	4.714
Adaptive	T10_ActionDensity	1.502	0.054	27.589	0.001	1.267	1.736	23.666
Adaptive	T11_InputStability	-0.546	0.027	-20.163	0.002	-0.662	-0.429	5.851
Adaptive	T12_ComboSuccessRate	0.589	0.041	14.375	0.005	0.413	0.765	13.414
Adaptive	T13_IdleTimeRatio	0.422	0.031	13.744	0.005	0.290	0.555	7.542
Adaptive	T14_ErrorRate	-1.257	0.040	-31.210	0.001	-1.430	-1.084	12.956
Adaptive	T15_SessionDuration	-1.085	0.036	-30.079	0.001	-1.240	-0.930	10.390

Table S11. Questionnaire Flow by group (Mann–Whitney U, rank-biserial r)

scale	rule_median	adaptive_median	U	p	rank_biserial_r
Flow (24 items)	3.833	3.083	420.000	0.000	0.736
Concentration	4.125	3.250	368.000	0.003	0.521
Challenge-skill balance	3.875	3.250	341.500	0.020	0.411
Control / system response	3.875	2.875	373.000	0.002	0.541
Enjoyment / motivation	4.500	3.250	420.500	0.000	0.738
Time / deep engagement	4.000	3.000	386.500	0.001	0.597
Pressure / flow experience	3.750	2.250	418.000	0.000	0.727